

Ahsanullah University of Science & Technology

Department of Computer Science & Engineering

Term Assignment

Course No : CSE 3208

Course Title : Introduction to Artificial Intelligence Lab

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Introduction

Diabetes is a chronic medical condition that affects millions of people worldwide. Early detection and prediction of diabetes can significantly improve patient outcomes by enabling timely medical intervention. With the advancement of machine learning, predictive models can be developed to assist in diagnosing diseases based on patient data.

The objective of this project is to build and compare multiple machine learning models to predict whether a patient has diabetes based on numerical health-related features. Three classification algorithms—Decision Tree, Naïve Bayes, and K-Nearest Neighbors (KNN)—are implemented and evaluated using standard performance metrics.

Dataset Description

The dataset used in this project is the **Diabetes Prediction Dataset**, which contains patient-related health information. The dataset includes both categorical and numerical features such as:

- Age
- Hypertension
- Heart disease
- BMI (Body Mass Index)
- HbA1c level
- Blood glucose level
- Gender (categorical)
- Smoking history (categorical)
- Diabetes (target variable)

The target variable '**diabetes**' is binary:

- 0 → No Diabetes
- 1 → Diabetes

For this project, only numerical features were considered to simplify preprocessing and model training.

Data Preprocessing

To ensure the models were trained on clean, high-quality data, the following steps were performed:

- **Feature Selection:** Categorical features (gender and smoking history) were removed to satisfy the "numeric-only" requirement of the assignment.
- **Missing Value Handling:** Rows containing **NaN** values were dropped to prevent training errors.
- **Data Splitting:** The data was split using a 80/20 ratio for training and testing. **Stratification** was applied to maintain the same ratio of diabetic vs. non-diabetic cases in both sets.
- **Feature Scaling:** StandardScaler was applied to normalize the features, ensuring that variables with larger ranges (like glucose levels) do not disproportionately influence the KNN model.

Methods

Three supervised machine learning algorithms were used:

1. Decision Tree Classifier

- A tree-based model that splits data based on feature values.
- Configured with `max_depth = 5` to prevent overfitting.
- Advantages: Easy to interpret and handle non-linear relationships.

2. Naïve Bayes (GaussianNB)

- A probabilistic classifier based on Bayes' theorem.
- Assumes independence between features.
- Efficient and works well with small datasets.

3. K-Nearest Neighbors (KNN)

- A distance-based algorithm that classifies based on nearest neighbors.
- Configured with `k = 5`.
- Sensitive to feature scaling, hence normalization was necessary.

4. Evaluation Metrics

The models were evaluated using:

- **Accuracy:** Overall correctness
- **Precision:** Correct positive predictions
- **Recall:** Ability to detect actual positives
- **F1-Score:** Balance between precision and recall

Confusion matrices were also used to visualize classification performance.

Results and Analysis

Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.9723	1.0000	0.6741	0.8053
Naïve Bayes	0.9049	0.4583	0.6559	0.5396
KNN	0.9654	0.9078	0.6600	0.7643

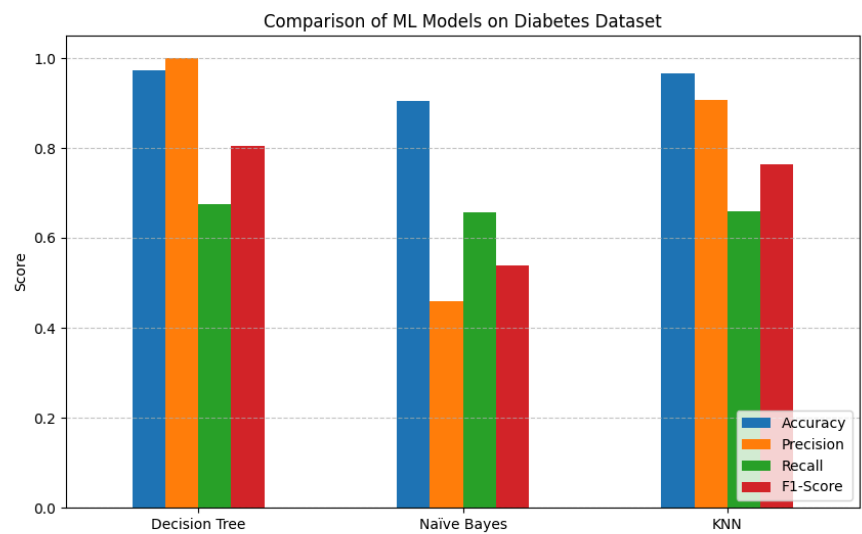


Fig 1: Comparison of ML Models on Diabetes Dataset

Analysis

- **Decision Tree**

- Achieved the highest accuracy (97.23%) and perfect precision (1.0).
- However, recall is relatively lower (0.6741), meaning it misses some diabetic cases.
- High precision indicates very few false positives.

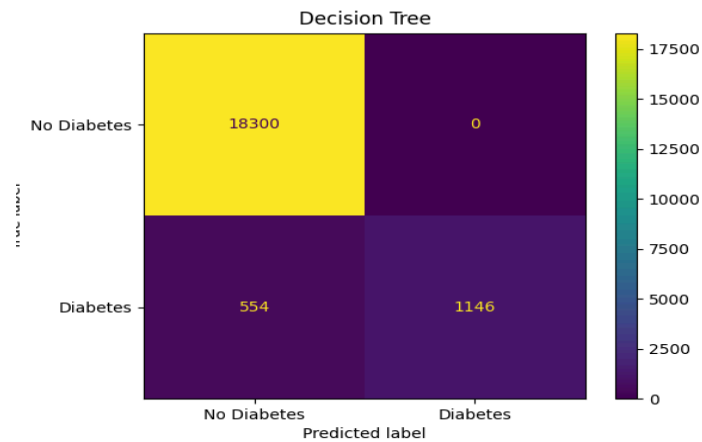


Fig 2: Confusion Matrix of Decision Tree Model.

- **Naïve Bayes**

- Lowest performance among all models.
- Particularly weak precision (0.4583), meaning many false positives.
- Moderate recall suggests it detects some diabetic cases but not reliably.

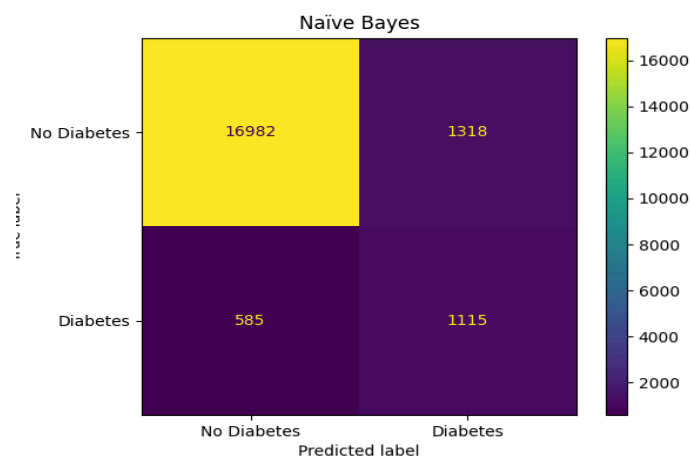


Fig 3: Confusion Matrix of Naïve Bayes Model.

- **KNN**

- Balanced performance across all metrics.
- High precision (0.9078) and good accuracy (96.54%).
- Slightly lower recall compared to Decision Tree.

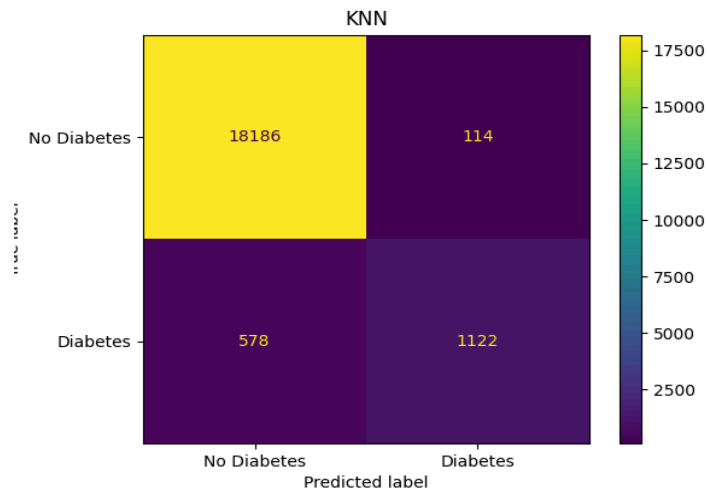


Fig 4: Confusion Matrix of KNN Model.

Key Observations

- The dataset may be imbalanced, which affects recall scores.
- High accuracy alone is not sufficient; recall is crucial in medical diagnosis.
- The Decision Tree is strong but may miss actual diabetic patients.
- KNN provides a better balance between precision and recall.

Conclusion

In this project, three machine learning models were implemented to predict diabetes using numerical health data. Among the models:

- **Decision Tree** achieved the highest accuracy and precision but had moderate recall.
- **KNN** provides a balanced performance and is a strong alternative.
- **Naïve Bayes** performed the weakest overall.

For medical applications, recall is especially important, as failing to detect a disease can have serious consequences. Therefore, while Decision Tree performs best in accuracy, KNN may be more practical due to its balanced behavior.

Future improvements could involve "Oversampling" techniques to help the models better recognize the minority (diabetic) class.